FOUNDATIONS OF ARTIFICIAL INTELLIGENCE PROJECT

Scribe AI

Medical SOAP Notes Generator

AI-assisted clinical documentation from consultation audio to structured SOAP notes.

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VOICE → TRANSCRIPT → SOAP NOTE

Problem Statement

Why clinical documentation needs a more efficient workflow

Time-consuming charting

Clinicians must document consultations while balancing direct patient care and other clinical responsibilities.

Unstructured information

Spoken information can be lengthy, repetitive, or difficult to convert consistently into a standard clinical format.

Documentation burden

Manual entry can contribute to delays, inconsistent notes, and reduced time available for patient interaction.

Project objective

Develop an AI-assisted system that converts consultation audio into editable, structured SOAP notes for clinician review.

Abstract

Scribe AI: Medical SOAP Notes Generator

Scribe AI is a clinical documentation assistant designed to reduce the manual effort involved in preparing consultation notes. The proposed workflow accepts recorded doctor–patient conversations, converts speech into text, and applies natural language processing to organize relevant information into the four sections of a SOAP note: Subjective, Objective, Assessment, and Plan. The generated note is presented for clinician review and editing before it is saved or exported. The system combines a web interface, an API layer, speech recognition, NLP-based information extraction, and persistent storage. It is intended to support documentation efficiency—not to replace clinical judgement or independently make diagnoses.

Input

Consultation audio

Processing

Speech-to-text + NLP

Output

Editable SOAP note

Proposed Solution

A human-in-the-loop pipeline for clinical note generation

1. Capture:
The doctor records or uploads the consultation audio.

2. Transcribe:
AI converts the audio into a written transcript.

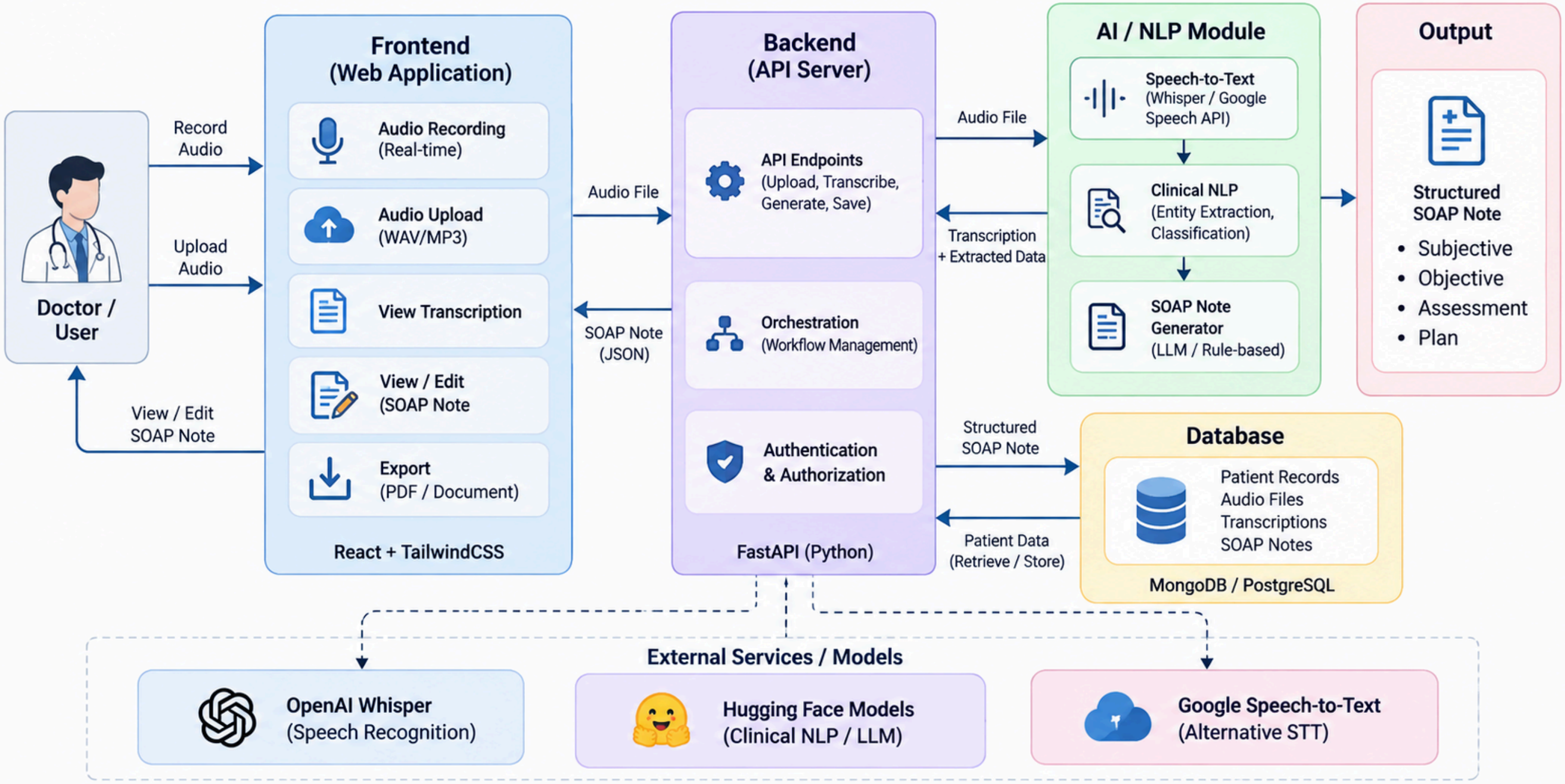
3. Extract:
NLP identifies relevant symptoms, medical findings, assessments, and treatment plans.

4. Structure:
The extracted information is organized into Subjective, Objective, Assessment, and Plan sections.

5. Review & save:
The doctor reviews, edits, and approves the generated notes before saving or exporting them.

Key safeguard: AI-generated notes remain drafts until reviewed by a qualified clinician.

System Architecture



Technology Stack

Candidate technologies reflected in the project concept

Frontend

React, HTML, CSS, JavaScript
Responsive interface for audio capture, transcript viewing, and note editing.

Backend API

Python + FastAPI (proposed)
REST endpoints to coordinate transcription, NLP processing, and persistence.

Speech & NLP

OpenAI Whisper or a speech-recognition API; spaCy or a BERT-based model for clinical information extraction.

Database & Export

PostgreSQL or MongoDB for storing approved notes and metadata; document export can be added as required.

Feasibility & Scalability

Practical considerations for a staged implementation

Technical feasibility

Built from established web, API, speech-recognition, NLP, and database components. A prototype can be developed incrementally.

Operational feasibility

Fits a clinician review workflow: record, transcribe, generate, edit, approve, and save.

Scalability

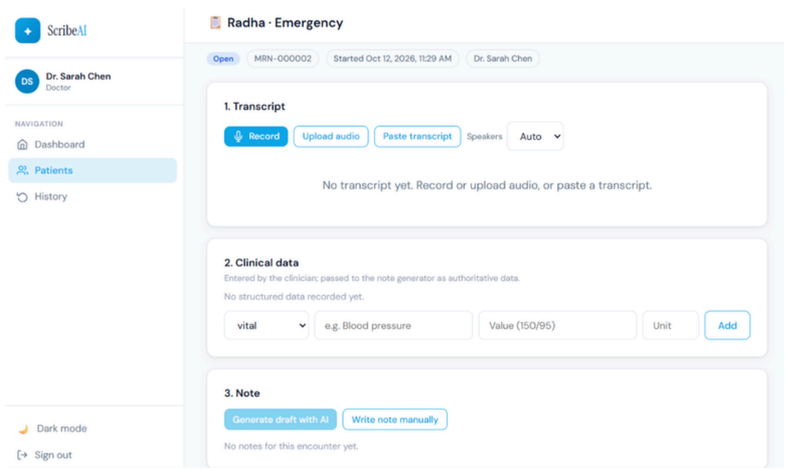
Separate API and AI-processing components; queue longer jobs, deploy services independently, and scale compute/storage as usage grows.

Risks and mitigations

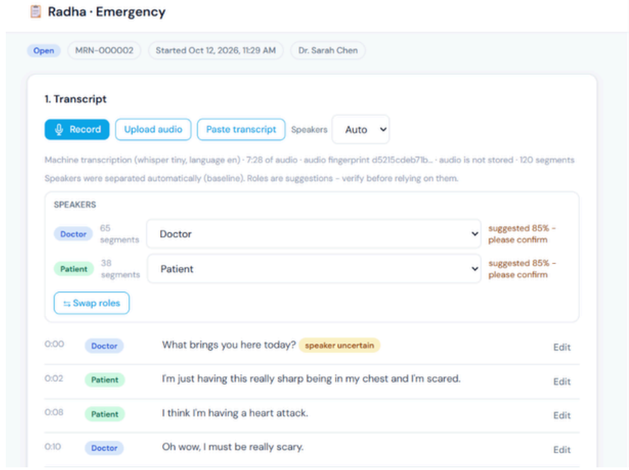
Audio quality and accents can affect transcription → test on varied, consented recordings.
Clinical terms may be misrecognized → retain transcript visibility and clinician editing.
Sensitive patient data → use authentication, encryption, least-privilege access, retention rules, and appropriate compliance review.
Performance claims must be validated → measure latency, transcription quality, note completeness, and user effort.

Results & Screenshots

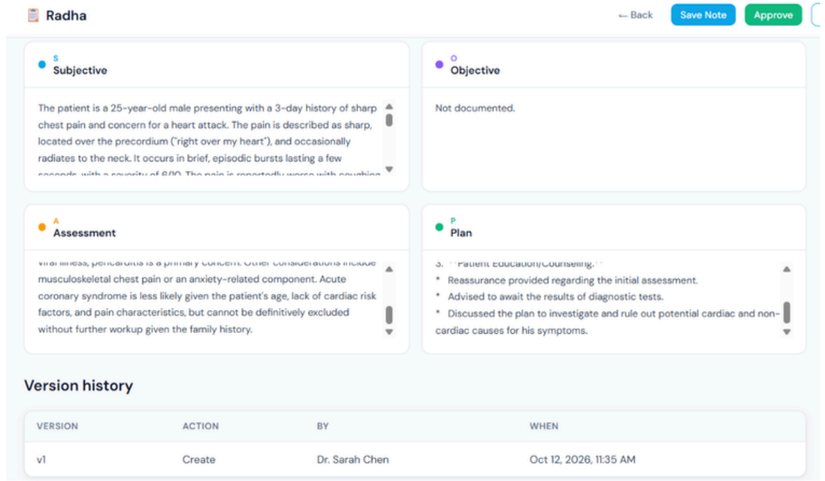
Insert actual application screenshots and measured results here



1. Audio / Upload Screen



2. Transcript / Processing



3. Generated SOAP Note

Literature Survey

Research themes relevant to speech-based clinical documentation

Theme	Key finding / relevance	Implication for Scribe AI
Clinical documentation burden	Ambient documentation tools aim to reduce time spent on manual note entry and support clinician workflow.	Evaluate time saved and clinician workload.
Automatic speech recognition	Medical speech can include specialized vocabulary, accents, interruptions, and background noise.	Measure word/error rates on representative audio.
Clinical NLP	Entity extraction and language models can help map free text into structured clinical concepts.	Check completeness, omissions, and hallucinated details.
Human oversight	Clinical decision support requires review and careful handling of errors and uncertainty.	Keep the clinician in control of final approval.

References

Starter references to verify and format according to your institution’s citation style

1. SOAP documentation format: Subjective, Objective, Assessment, and Plan — standard clinical documentation framework.
2. OpenAI. Whisper: Robust Speech Recognition via Large-Scale Weak Supervision. arXiv:2212.04356, 2022.
3. Honnibal, M. & Montani, I. spaCy 2: Natural language understanding with Bloom embeddings, convolutional neural networks and incremental parsing. 2017.
4. Devlin, J., Chang, M.-W., Lee, K., & Toutanova, K. BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding. NAACL-HLT, 2019.
5. FastAPI documentation. https://fastapi.tiangolo.com/
6. React documentation. https://react.dev/
7. PostgreSQL documentation. https://www.postgresql.org/docs/ (if PostgreSQL is selected)